Higher National Diploma in Accountancy
First Year, First Semester Examination – 2023
DA 1123 – Business Mathematics

Instructions for candidates:
Answer five (05) questions **including question number 01**
Calculators are allowed.

No. of questions : 06
No. of pages : 04
Time : **Three(3) hours**

Question 01

Write the correct answer.

(i) The numbers that have more than two factors are defined as:

(*a*) Even (*b*) Cardinal (*c*) Composite (*d*) Prime

(ii) Which of the following number is an integer?

(*a*) $2\sqrt{2}$ (*b*) 0.5 (*c*) $2\sqrt{9}$ (*d*) 0.125

(iii) Vihara says all the fractions are rational numbers and Minuki says all the rational numbers are fractions. Who is correct?

(*a*) Vihara (*b*) Minuki (*c*) Both of them (*d*) None of them

(iv) Which of the following number is not real?

(*a*) – 11 (*b*) 0.125 (*c*) $\sqrt{-4}$ (*d*) π

(v) The value of $[8]^{-\frac{2}{3}}$ is equal to:

(*a*) $-\frac{1}{4}$ (*b*) $\frac{1}{4}$ (*c*) 4 (*d*) – 4

(vi) If $\frac{(25)^{7.5} \times (5)^{2.5}}{(125)^{1.5}} = 5^m$, then the value of *m* is:

(*a*) 9.5 (*b*) 12 (*c*) 13 (*d*) 8.5

(vii) If $\left[\frac{3}{2}\right]^{(x-1)} = \left[\frac{2}{3}\right]^{(x-3)}$, then the value of *x* is equal to:

(*a*) 1 (*b*) 2 (*c*) – 1 (*d*) – 2

(viii) $[16\,x^{12}]^{\frac{3}{4}}$ is equal to:

(*a*) $16x^9$ (*b*) $8x^9$ (*c*) $9x^8$ (*d*) $16x^8$

(ix) $\frac{6a^3c^2}{b^2} \times \frac{b^3a^0}{2c^2}$ is equal to:

(*a*) $3a^3b$ (*b*) $3a^4b$ (*c*) $3a^4b^{-1}$ (*d*) $3a^3b^{-1}$

(x) One of the factors of the polynomial $x^3 - 9x^2 + 14x$ is:

(*a*) $(x + 7)$ (*b*) $(x + 2)$ (*c*) $(x - 2)$ (*d*) $(x + 3)$

(xi) Factors of $x^4 - 1$ are:

(*a*) $(x - 1)(x + 1)$ (*b*) $(x + 1)(x + 1)$ (*c*) $(x^2 + 1)(x^2 + 1)$ (*d*) $(x^2 + 1)(x^2 - 1)$

(xii) Anil takes a loan of Rs.18500 to buy a phone at the rate of 10 % simple interest. The expected interest at the end of 4th year is:

(*a*) Rs.25900 (*b*) Rs.7400 (*c*) Rs.8585.85 (*d*) Rs.27085.85

(xiii) A person hopes to double the amount for a period of 10 years at a simple interest rate. The expected interest rate is:

(*a*) 12% (*b*) 10% (*c*) 8% (*d*) 6%

(xiv) Nimal buys a shirt marked at Rs.800. He receives a 10% discount. How much does he pay for the shirt?

(*a*) Rs.80 (*b*) Rs.700 (*c*) Rs.880 (*d*) Rs.720

(xv) A person invests Rs. 5000 at an 8% rate of compound interest for a period of 6 months. When the interest is calculated quarterly, expected total interest of this investment is:

(*a*) Rs.5412 (*b*) Rs.412 (*c*) Rs.5202 (*d*) Rs.202

(xvi) At what rate of compound interest per annum will a sum of Rs. 1200 become Rs. 1348.32 in two years?

(*a*) 6% (*b*) 6.5% (*c*) 7% (*d*) 7.5%

(xvii) Ruwan is going to take a loan of Rs.500000 and agreed to pay the loan with interest at 6% per annum within 4 years. The monthly installment of this is:

(*a*) Rs.17142.51 (*b*) Rs.11742.51 (*c*) Rs.14172.51 (*d*) Rs.12417.51

(xviii) 30 boys and 25 girls, in a class of 55 students, were given a test. The mean mark for boys was 58. The mean mark for the girls was 69. Then the mean mark of all 55 students is:

(*a*) 63.5 (*b*) 63 (*c*) 60 (*d*) 61

(xix) The mode of the observations 3, 2, 7, 6, 5, x and 9 is 6. The value of x is equal to:

(*a*) 4 (*b*) 9 (*c*) 6 (*d*) None of these

(xx) The mean of the observations 3, 2, 7, 6, 5, x and 9 is 6. The value of x is equal to:

(*a*) 4 (*b*) 9 (*c*) 6 (*d*) None of these

(2×20 = 40 marks)

[Total 40 marks]

Question 02

(i) Solve $xy = 24$ and $x = y - 5$

(4 marks)

(ii) A person appointed for a post gets a monthly salary of Rs. 44200 during the fourth year of his service. His monthly salary during the 10th year of his service is Rs. 46600. Find;

(*a*) the initial salary of that post. **(2 marks)**

(*b*) the annual increment of his salary. **(2 marks)**

(iii) If $y = x^3 - 2x^2 + 7x + 5$ then find the followings:

(*a*) $\frac{dy}{dx}$ **(3 marks)**

(*b*) $\frac{d^2y}{dx^2}$ **(2 marks)**

(*c*) x such that $\frac{d^2y}{dx^2} = 0$ **(2 marks)**

[Total 15 marks]

Question 03

In a certain production the cost function is $2500 + 4x + 0.01x^2$ and demand function is $31 - 0.01x$, where x is the production level. For this production, find;

(*a*) the revenue function **(2 marks)**

(*b*) the profit function **(2 marks)**

(*c*) number of units sold at the break-even points **(3 marks)**

(*d*) marginal cost (MC) and marginal revenue (MR) functions **(4 marks)**

(*e*) the production level that maximizes the profit **(2 marks)**

(*d*) maximum profit **(2 marks)**

[Total 15 marks]

Question 04

(i) ABC company is considering two projects X and Y. For each project the rate of return is 10% and the initial investment amount is Rs.100000. Both the projects are four-year projects and cash flows as follows:

Year	1	2	3	4
Project X	40000	50000	30000	10000
Project Y	30000	40000	40000	20000

For the above projects, find the followings;

(*a*) Payback period for each project. **(3 marks)**

(*b*) Net present value (NPV) for project X and Y **(3 marks)**

(*c*) The best project (based on the net present values) **(3 marks)**

(*b*) Internal rate of return (IRR) for the project Y. **(3 marks)**

(ii) Demand and supply equations in a production as follows:

Quantity of demand $Q_d = 4500 - 6p$ and

Quantity of demand $Q_s = -2300 + 4p$

Calculate the equilibrium price and quantity **(3 marks)**

[Total 15 marks]

Question 05

(i) Nimal is going to buy a camera on installments. He has to pay 7 annual installments each is Rs.35,000 right from the date of purchase. If the rate of compound interest is 16% then find the price of the camera. **(5 marks)**

(ii) Amith wants to collect Rs.45,300 at the beginning of the 10th year. For that, he is going to invest a fixed amount at the beginning of each quarter at 8% rate (per annum) of compound interest. Find the installment. **(5 marks)**

(iii) A person invests Rs. 50000 at an 8% rate of compound interest for a period of 5 years. Find the total amount of this investment, at the end of 5th year, when the interest is calculated quarterly.

(5 marks)

[Total 15 marks]

Question 06

(i) Income of a factory decreases every year by 8% of the income it earned in the previous year. The income in the first year was Rs. 120000. Find the income of the factory in the sixth year.

(5 marks)

(ii) In February, Sara begins to save money for a Computer. She starts with Rs.2000. Each month she plans to save 10% than the previous month. How much money will she save in October?

(5 marks)

(iii) Anura begins to save money for a computer which costs Rs. 78000. He starts with Rs. 3000. Each month he plans to save 10% than the previous month. How long (in months) will it take to collect the total amount for the computer? **(5 marks)**

[Total 15 marks]